DATA ANALYTICS
PORTFOLIO REPORT

Customer Churn Analysis & Customer Intelligence

An independent professional review of the supplied churn-analysis project, its reported findings, KPI definitions, data model, and recommended validation steps.

Project domain | OTT / Subscription Analytics

Core tools | SQL · Python · pandas · NumPy · SQLite · Matplotlib · Seaborn

Prepared: 25 October 2026

Executive Summary

This report reframes the supplied churn-analysis brief into a more professional, decision-oriented portfolio document. The original brief describes a relational analytics workflow across customer, subscription, and support data, and reports several headline metrics. The analysis below distinguishes **figures reported in the source** from findings that still require verification against the underlying dataset.

28.6%	71.4%	55.6%	8.3%
Reported overall churn rate	Reported retention rate	Reported monthly-contract churn	Reported annual-contract churn

Headline interpretation. The source reports a substantially higher churn rate among monthly-contract subscribers than annual-contract subscribers. This is a useful retention hypothesis, but the underlying customer records, churn labels, time window, and denominators were not included in the supplied report; therefore, the figures should be described as *reported results* until independently reproduced.

Overall assessment

Strengths	Priority improvements
Clear business question; relational data model; practical KPIs; segmentation; action-oriented recommendations.	Reproducible calculations; consistent currency and units; validated joins; clear metric definitions; evidence for causal claims.

Important limitation: This document reviews the supplied PDF only. It does not claim to have rerun SQL queries, inspected raw customer-level records, or independently verified the reported metrics.

1. Business Problem and Analytical Objectives

Customer churn occurs when a customer stops using or paying for a service. In a subscription business, churn affects recurring revenue, customer lifetime value, and the efficiency of acquisition spending. The original project focuses on an OTT-style subscription platform and proposes answering three questions:

- **Who is leaving?** Identify customer segments with elevated churn, including contract type, plan, geography, and customer characteristics.
- **What may be associated with churn?** Explore cancellation reasons, support escalations, customer satisfaction, and subscription characteristics.
- **When does risk appear?** Examine churn by month and tenure to identify periods where retention interventions may be most useful.

Business objectives

1. Establish a reliable baseline for churn and retention.
2. Compare churn across subscription plans and contract types.
3. quantify recurring revenue associated with churned customers and customers flagged as high risk.
4. Investigate whether support issues and low satisfaction co-occur with churn.
5. Translate validated findings into prioritized retention actions.

Scope note

The source brief lists a broad multi-industry use case, but the actual example and metrics are subscription/OTT-oriented. Churn definitions should be tailored to the business model and analysis period rather than reused unchanged across industries.

2. Data Model and Preparation

The source describes three tables in a database named `customer_churn`. The proposed join key is `customerid`.

Table	Fields described in the source	Analytical use
`db_customer`	customerid, name, country, state, gender, dob, interests, pincode	Customer profile and geographic segmentation
`db_subscription`	customerid, subscription_start_date, subscription_type, renewal_date, plan_type, contract_type, cancellation_date, cancellation_reason, monthly_charges, cltv, churn_score	Subscription lifecycle, churn status, plan and revenue analysis
`db_support`	customerid, complaint_date, Escalations, csat_score, col_1, comment	Support interaction and satisfaction analysis

Critical data-quality checks before analysis

- Confirm whether `customerid` is unique in the customer table and determine whether subscription/support tables contain multiple records per customer.
- Define a single, explicit churn flag and analysis cut-off date. A cancellation date, an inactive membership, and a churn score are not interchangeable.
- Check date parsing, impossible dates, missing values, duplicate rows, negative charges, and inconsistent category labels.
- When joining support records to subscriptions, aggregate support data to the required grain first or use distinct customer-level logic to avoid duplicating charges.
- Confirm the meaning of `col_1` and the unit/scale of CLTV and churn_score before using them.

3. KPI Framework and Definitions

The original report lists the following KPIs. The definitions below clarify how they should be operationalized for a defensible analysis. Use one consistent reporting period and document the population included in each denominator.

KPI	Recommended definition	Validation note
Churn rate	Customers who churned during the period ÷ customers eligible to churn during the period × 100	Define period and eligible population; avoid mixing historical churn with current customer count.
Retention rate	1 – churn rate, only when the churn and retention populations and period are aligned	May not equal a cohort retention metric; state the chosen definition.
Churn by plan / contract / state	Churn rate calculated separately for each segment	Compare rates, not just raw churn counts; show segment size.
ARPU	Recurring revenue for the period ÷ average or clearly defined active-customer denominator	Use a consistent currency and time unit; distinguish ARPU from average charge per account.
Average tenure	Average elapsed time from subscription start to cancellation for churned customers; use a fixed cut-off for active customers	Report active and churned tenure separately if relevant.
Revenue at risk	Sum of recurring charges for customers classified as at risk, under a documented risk rule	This is exposure, not realized revenue loss; avoid double-counting churned customers.
Escalation rate	Escalated support cases ÷ total support cases × 100	Do not divide escalation count by complaint count unless units and case structure are clear.
Complaints per customer	Number of complaint records ÷ distinct customers in the defined population	State whether this is all customers or customers who complained.
Escalations and churn	Compare churn rates for customers with ≥1 escalation versus those with none	Association does not establish causation; adjust for tenure, plan and other confounders if possible.

4. Findings Reported in the Source

The figures in this section are transcribed from pages 7–9 of the supplied report. They are not independently reproduced because the underlying dataset and executable analysis were not supplied.

Reported result	Value in source	Interpretation / caveat
Overall churn / retention	28.6% / 71.4%	The two figures sum to 100%; denominator and reporting period still need to be confirmed.
Monthly vs annual contract churn	55.6% vs 8.3%	Monthly-contract churn is reported as about 6.7× annual-contract churn. Validate sample sizes and definitions before making a strong claim.
Geographic pattern	Karnataka reported as most affected state	Investigate possible pricing, service, technical or cohort effects; the report does not establish the cause.
Timing pattern	Most churn reportedly occurred in September 2024	Confirm that the dataset covers comparable months and the same churn event definition.
Average tenure / ARPU	1,451 days / Rs 18.8	Units and currency need confirmation; ARPU denominator and period should be documented.
Revenue / revenue loss	Total revenue = 395; loss = 74; revenue loss = 18%	Currency and aggregation period are not stated clearly in the source.
CLTV lost / at-risk customers	CLTV lost = 2,047; six at-risk customers mentioned	The source later reports $73.94/mo MRR leakage. Confirm currency, unit, cohort, and whether figures reconcile.

Key analytical insight

The contract-type gap is the most decision-relevant headline in the report: monthly-contract subscribers are reported to churn at a much higher rate than annual-contract subscribers. A responsible next step is to investigate whether the gap persists after accounting for tenure, plan, customer mix, and cohort timing. Annual-contract customers may differ systematically from monthly customers, so the gap alone does not prove that migrating customers to annual plans will reduce churn.

Reported recommendations to validate

- Investigate churn concentration in Karnataka and September 2024.
- Review any changes to pricing, plan design, service reliability, or content around the reported churn period.
- Review customers with medium/high churn scores, prioritizing by verified customer value and contact permissions.
- Analyze cancellation reasons and support experiences before proposing targeted interventions.

5. Gaps to Resolve Before Publishing

Issue	Why it matters	Required correction
No raw data / code included	Metrics cannot be reproduced from the PDF alone.	Provide dataset, SQL queries, Python notebook and a metric dictionary.
Currency inconsistency	The source uses Rs and $ in different sections.	Declare source currency; convert only with a documented rate and date if necessary.
Ambiguous revenue labels	“Total revenue = 395” lacks unit and time window.	Specify currency, period, whether revenue is monthly/annual, and aggregation grain.
Risk score treated as revenue loss	A churn score above 70 indicates risk only if the score is defined and calibrated.	Report exposure separately from realized churned MRR; validate score thresholds.
Possible join duplication	Multiple support cases per customer can inflate charges after joining tables.	Aggregate support to customer or period before joining; reconcile totals before and after joins.
Causal language	Observed relationships do not prove that support or contract type causes churn.	Use “associated with” unless supported by an appropriate causal design or experiment.
“20+ KPIs” claim	The source mentions 20+ KPIs but lists only a subset in the report.	List the actual KPIs and provide definitions or remove the unsupported count.

6. Recommended Implementation Roadmap

Stage	Work	Deliverable / quality gate
1. Audit	Inspect schema, row counts, keys, nulls, date ranges, categories and duplicates.	Data dictionary + quality report
2. Extract	Use SQL to join the tables at a defined grain and extract analysis-ready data to Python.	Reproducible SQL + reconciled row counts
3. Engineer	Create churn flag, tenure, contract/plan segments, support metrics and revenue fields.	Documented feature definitions
4. Analyze	Calculate overall and segment churn; compare cohorts; test support/churn associations.	Validated KPI tables + uncertainty notes
5. Visualize	Create clear trend, segment, cancellation-reason and support charts.	Executive dashboard / figures
6. Recommend	Prioritize actions by expected value, feasibility and evidence strength.	Action plan with owner, metric and review date

7. Recommended Business Actions

Actions should be prioritized after validating the underlying numbers. The recommendations below are hypotheses to test, not claims that the reported causes have already been established.

Priority	Action	Success measure
1 — High	Investigate monthly-contract churn. Segment by plan, tenure, acquisition cohort, price changes and cancellation reason.	Monthly-contract churn rate and retention by cohort
2 — High	Review high-risk customers using a verified risk score and customer value; test relevant retention offers or service recovery.	Incremental retention vs. a control group; offer cost and retained MRR
3 — High	Analyze support escalations and CSAT before cancellation, controlling for customer tenure and plan where possible.	Churn rate by support experience; resolution time; CSAT
4 — Medium	Investigate the reported Karnataka and September 2024 patterns for product, pricing, technical or data-quality explanations.	Validated segment trends and documented root-cause evidence
5 — Medium	Use cancellation reasons to guide product, pricing and content reviews.	Reason-specific churn and post-intervention change

8. Portfolio Presentation and CV Guidance

A professional portfolio version should include evidence of both technical execution and business judgment. Suggested repository structure:

```
customer-churn-analysis/
■■■ data/ (sample or appropriately anonymized data)
■■■ sql/ (extraction and aggregation queries)
■■■ notebooks/ (cleaning, EDA, KPI calculations)
■■■ outputs/ (charts and KPI tables)
■■■ reports/ (final PDF and slide deck)
■■■ README.md
■■■ requirements.txt
```

Suggested CV bullet — use only after reproducing the results

> Built an end-to-end customer churn analytics workflow using SQL and Python to integrate customer, subscription and support data; engineered churn and revenue KPIs, compared retention across contract segments, and translated validated findings into targeted retention recommendations.

Avoid claiming a quantified business impact, live-data analysis, a deployed model, or a causal reduction in churn unless you personally performed and can demonstrate that work.

9. Conclusion

The supplied project brief is a useful starting point for a Data Analyst portfolio because it links relational data, KPI engineering, segmentation and business recommendations. Its strongest reported signal is the difference between monthly- and annual-contract churn. To make the project credible and interview-ready, the next step is to reproduce the metrics from the underlying data, document definitions and denominators, reconcile currencies and revenue figures, and present recommendations as testable hypotheses backed by evidence.

Source and limitations

Primary source: *Data Analytics Project – Churn Analysis Report.pdf*, supplied by the user (11 pages). The project description, schema, KPI list and reported numerical findings in this document are based on that source. Analytical recommendations and validation notes are independent review commentary. No raw dataset, SQL script, Python notebook, or dashboard was provided for this review.